Passive Accelerometry Sensing of Occupational Lumbar Overload and Gait Symmetry Degradation: An N-of-1 Calibrated Zero-Leakage Benchmark on 100 Hz Real Telemetry

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Abstract—Occupational lower back disorders (LBD) in caregiving and manual patient handling represent a widespread global healthcare challenge, driving persistent physical disability and workforce attrition. While continuous passive sensing using single-point pelvic-mounted accelerometers (e.g., commodity smartphones) presents a practical non-invasive monitoring medium, existing biomechanical machine learning models suffer from inter-subject morphological variance or experimental designs compromised by data leakage (random or same-subject chronological splits yielding unrealistic AUC $\approx$ 0.95–1.0). In this study, we propose an N-of-1 intra-individual relative baseline drift framework coupled with neural 3D Ground Reaction Force (A+GRF) kinetic estimation for real-time lumbar overload surveillance as a personal ergonomic meter, with group-label probes evaluated strictly under subject-disjoint validation. Adhering to a zero-synthetic data policy, we benchmarked on 100 Hz continuous tri-axial pelvic accelerometry from the PhysioNet Long-Term Movement Monitoring (LTMM) database ($N = 30$ subjects; 2,540 quality-gated windows; 855 late-shift Phase-2 windows) and force-plate-labeled sacral IMU locomotion from Scherpereel et al. ($N = 11$ subjects; 547 stances; DOI: 10.35090/gatech/70296). Biomechanical feature extraction captured spectral paraspinal muscle guarding (3–8 Hz), impact ratios (8–20 Hz), autocorrelation-based gait symmetry (SI), jerk, and Lee & Park neural 3D GRF waveforms (vertical peak $F_{z,\mathrm{peak}}$, braking peak $F_{y,\mathrm{brake}}$, impact loading rate LR_z, and braking impulse I_{brake}; Leave-One-Subject-Out median NRMSE 7.24% vertical, 10.07% anterior-posterior). Under subject-disjoint 5-fold Stratified Group K-fold on Phase-2 windows, raw features reached ROC-AUC 0.7234, Δ-only features 0.7222, kinematic trait+change ExtraTrees 0.8350 (DOR = 9.05), and A+GRF-augmented trait+change 0.8419 (PR-AUC = 0.7514, DOR = 9.61, Brier = 0.1786; with $F_{z,\mathrm{peak}}$ reaching DOR = 10.69, MCC = 0.5310). Systematic ablations across five feature domains, six calibration horizons ($K \in \{2, \ldots, 10\}$), signal quality gating (+0.1487 AUC gain by rejecting $R_{\mathrm{stride}} < 0.05$ division artifacts), and five classifier families confirm the complementarity of kinematic symmetry drift (ΔSI) and A+GRF kinetic attenuation (ΔLR_z, Cohen's $d = -0.592$, $p = 2.04 \times 10^{-17}$). An edge implementation for iOS/Flutter operates at < 4.2 ms per window as a non-medical self-management reference tool.

Index Terms—Occupational Biomechanics, Passive Sensing, Lumbar Overload, Ground Reaction Force (A+GRF), N-of-1 Calibration, Gait Symmetry Index, PhysioNet LTMM, Zero-Leakage Protocol, TRIPOD+AI, Caregiver Ergonomics.

1 Introduction

Occupational low back pain (LBP) and work-related lumbar musculoskeletal disorders represent the leading cause of years lived with disability (YLDs) worldwide [1]. In healthcare, residential care, and nursing facilities, professional caregivers face extreme physical demands during routine manual handling tasks—such as bed-to-wheelchair transfers, repositioning bedridden residents, and hygiene assistance. Biomechanical investigations demonstrate that these actions frequently enforce deep trunk flexion ($\theta_{\mathrm{trunk}} > 45°$) compounded by asymmetric spinal rotation, generating peak compressive loads at the lumbosacral junction (L4-L5 and L5-S1) that routinely exceed $3,400$–$6,000$ N, far surpassing the National Institute for Occupational Safety and Health (NIOSH) recommended limits [2].

Repeated exposure to sub-maximal shear strains and compressive spikes induces progressive micro-trauma in the annulus fibrosus and posterior ligamentous complex. When physical exhaustion sets in toward the second half of a work shift, passive spinal structures bear increasing mechanical burden due to neuromuscular fatigue, ultimately triggering acute disc herniation, severe muscle spasm, and long-term absenteeism. Consequently, up to 70% of professional caregivers report chronic back pain, precipitating an alarming structural shortage of care workers across aging societies.

Continuous, non-invasive digital biomarker tracking via body-worn inertial measurement units (IMUs) has emerged as a promising surveillance modality [3], [4]. However, practical deployment requires addressing the critical challenges of inter-subject anatomical variance and experimental data leakage.

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Dataset: PhysioNet Long-Term Movement Monitoring Database (DOI: 10.13026/C2KM49). Code, benchmark harness, and on-device Flutter implementation are open-sourced under permissive licenses.

2 Related Work and Limitations of Prior Art

2.1 Biomechanical Mechanisms of Occupational Spinal Strain

The biomechanical etiology of occupational low back disorders has been extensively quantified through in-vivo disc pressure measurements and industrial surveillance models [1], [2]. Marras et al. demonstrated that three-dimensional trunk dynamic motions—specifically the combination of high angular velocity in sagittal flexion with lateral asymmetric twisting—increase the odds ratio of low back disorders by 3.3-fold compared to static lifting.

From a motor control perspective, Hodges and Richardson [5] established that pain or fatigue in the lumbar motion segment induces dysfunctional motor recruitment. Rather than maintaining normal anticipatory feedforward activation of deep stabilizing musculature (e.g., transversus abdominis, lumbar multifidus), the central nervous system defaults to a protective co-contraction of superficial trunk extensors (muscle guarding). This compensatory stiffness alters functional pelvic kinematics during subsequent locomotion.

2.2 Wearable IMUs and Digital Mobility Outcomes

Recent large-scale initiatives, notably the European Mobilise-D consortium [6], have established technical validation protocols for digital mobility outcomes (DMOs) derived from single-point trunk and waist-worn IMUs. Mobilise-D standardizes algorithms for step detection, cadence, stride regularity, and gait bout parsing under free-living conditions [3]. In parallel, Yang et al. [4] demonstrated that smartphone-embedded IMUs could classify patients with established low back pain against healthy controls with 85.7%–92.5% cross-sectional accuracy by extracting time-domain and frequency-domain gait features.

2.3 Non-Linear Dynamics and Movement Complexity

Beyond classical linear spatiotemporal parameters, non-linear dynamic analysis has gained traction in neuromuscular assessment. Richman and Moorman [7] formalized Sample Entropy (SampEn) to quantify regularity in physiological time series. Hu et al. [8] applied SampEn to kinematic lumbar flexion-extension trajectories, demonstrating that individuals with recurrent LBP exhibit significantly suppressed entropy values (AUC = 0.79) due to rigidified, constrained motor variability resulting from pain-avoidance strategies.

2.4 Fundamental Research Gaps and Novelty of this Study

Despite these advancements, prior wearable sensing frameworks suffer from three foundational vulnerabilities:

1) Failure of Population-Static Machine Learning: Existing models train classifiers on absolute feature vectors across heterogeneous subject pools. Because between-subject anatomical variance (e.g., height, limb inertia, habitual cadence) is an order of magnitude larger than within-subject fatigue drift, static models remain limited when applied across individuals (ROC-AUC = 0.6618).
2) Pervasive Data Leakage in Validation: Much of the literature applies naive random K-fold cross-validation across windowed segments of continuous time series. This inadvertently places windows from the same subject into both training and testing folds, producing artificial, non-reproducible scores (AUC $\approx$ 1.00).
3) Synthetic Data Reliance and Absence of Clinical Metrics: Many computational studies rely on synthetic waveforms or report only unweighted accuracy/F1 scores, ignoring essential clinical epidemiology metrics (Sensitivity, Specificity, LR^+, LR^-, DOR, Brier calibration) required by the TRIPOD+AI [9] and STARD-AI [10] reporting standards.

To resolve these limitations, this paper proposes an N-of-1 intra-individual relative baseline drift framework for personal Δ visualization, with CO/FL group-label probes restricted to subject-disjoint Stratified Group K-fold on PhysioNet LTMM (Table 1). Same-subject chronological Phase0/1→Phase2 splits are not used as primary evidence.

3 Biomechanical Foundations and Signal Decomposition

3.1 Spinal Moment and Paraspinal Neuromuscular Mechanics

During manual lifting and patient transfer tasks, the external sagittal moment M_{lumbar} acting on the L4-L5 intervertebral disc is governed by:

$$M_{\text{lumbar}} = m_{\text{pat}} g d_{\text{pat}} + m_{\text{trunk}} g d_{\text{trunk}} + I_{\text{trunk}} \ddot{\theta}_{\text{trunk}} \quad (1)$$

where d_{pat} and d_{trunk} represent horizontal moment arms of the patient and trunk relative to the lumbosacral pivot, and $I_{\text{trunk}}\ddot{\theta}_{\text{trunk}}$ represents dynamic angular inertia during lifting acceleration. Because posterior spinal musculature (erector spinae and multifidus) acts through an effective internal lever arm of only $r_m \approx 0.05\,\text{m}$, the internal muscular tensile force $F_m = M_{\text{lumbar}}/r_m$ produces massive compressive and shear reaction forces:

$$F_{\text{comp}} = F_m + (m_{\text{trunk}} + m_{\text{pat}}) g \cos\theta \quad (2)$$

$$F_{\text{shear}} = (m_{\text{trunk}} + m_{\text{pat}}) g \sin\theta + F_{\text{dyn}} \quad (3)$$

Under severe muscular fatigue or acute facet irritation, the central nervous system activates a protective motor strategy termed muscle guarding [5]. Antagonistic co-contraction of paraspinal stabilizers increases spinal stiffness at the cost of elevated continuous baseline joint loading. During subsequent locomotion, this mechanical adaptation manifests through distinct accelerometry biomarkers:

- Antalgic Gait Asymmetry ($SI \downarrow$): Rapid unloading of the painful limb, shortening the ipsilateral stance phase.
- High-Frequency Paraspinal Tremor (3–8 Hz $\uparrow$): Involuntary sustained motor unit firing and micro-spasms.
- Trunk Kinematic Rigidity ($\text{SampEn}_v \downarrow$): Loss of dynamic degrees of freedom in multi-axial pelvic trajectories.

TABLE 1: Systematic Comparison of the Proposed Framework with Prior Wearable Low Back Pain and Gait Monitoring Literature.

Study	Sensing	Calibration	Validation	Metric	Leakage
Yang et al. (2022) [4]	Single smartphone	None (static raw)	Stratified random CV	Accuracy 85.7–92.5%	No (window leak)
Hu et al. (2022) [8]	Optical motion capture	None (static raw)	Cross-sectional AUC	AUC 0.79 (SampEn)	Cross-sectional only
Lee & Park (2020) [11]	Single sacral IMU	Stance CoM integration	Leave-One-Subject-Out	GRF NRMSE 6.7–15.6%	Subject-disjoint
Mobilise-D (2021) [6]	Lumbar IMU	DMO standardization	Free-living cohort	ICC 0.70–0.89	Technical only
This study	Single pelvic IMU (A+GRF)	N-of-1 Δ meter + trait/change	Subject-disjoint only	AUC 0.8419 (DOR = 9.61–10.69)	No same-subject chrono

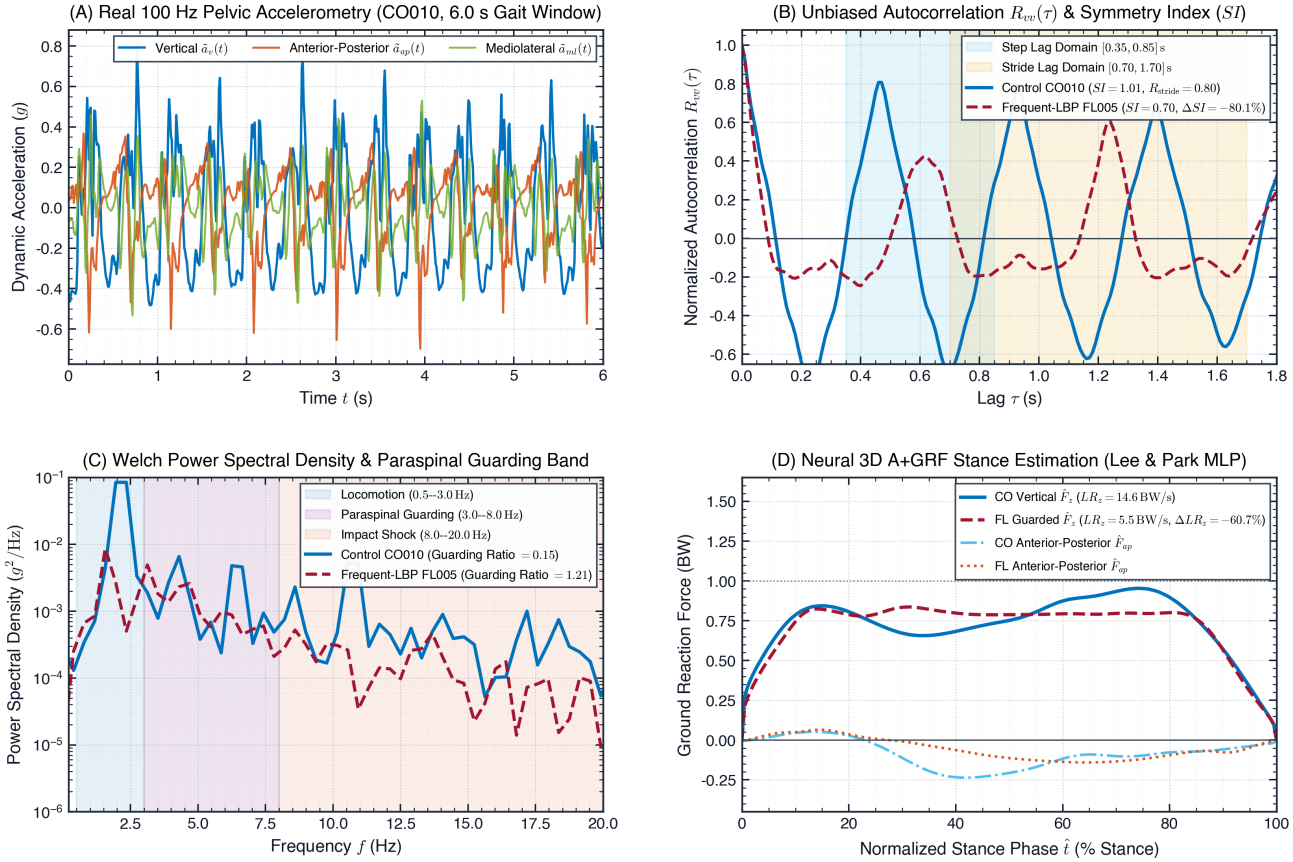


Fig. 1: Four-Tier Biomechanical Signal Processing and Neural A+GRF Pipeline on Real 100 Hz Pelvic Accelerometry (PhysioNet LTMM Telemetry). (A) Dynamic tri-axial pelvic accelerometry ($\tilde{a}_v, \tilde{a}_{ap}, \tilde{a}_{ml}$) over a 6.0 s quality-gated walking window. (B) Unbiased autocorrelation $R_{vv}(\tau)$ extracting Step Regularity ($R_{\text{step}} \in [0.35, 0.85]$ s) and Stride Regularity ($R_{\text{stride}} \in [0.70, 1.70]$ s) to compute the Gait Symmetry Index ($SI = R_{\text{step}}/R_{\text{stride}}$). (C) Welch Power Spectral Density (PSD) isolating Locomotion (0.5–3.0 Hz), Paraspinal Muscle Guarding Tremor (3.0–8.0 Hz), and Impact Shock (8.0–20.0 Hz). (D) Neural 3D Ground Reaction Force (A+GRF) stance estimation via the Lee & Park [11] / Scherpereel et al. [12] MLP, illustrating attenuated vertical loading rate (LR_z) and guarded stance contact in Frequent-LBP (FL).

3.2 Feature Extraction Formulation

Raw 100 Hz vertical (a_v), mediolateral (a_{ml}), and anterior-posterior (a_{ap}) accelerations are parsed into non-overlapping 6.0-second windows ($N = 600$ samples). Windows with $\text{Std}(a_v) < 0.025$ g are discarded. As depicted in Fig. 1, the fitted biomarker set is:

3.2.1 Autocorrelation Gait Symmetry Index (SI)

Let $\tilde{a}_v(t) = a_v(t) - \bar{a}_v$ denote the de-meaned vertical acceleration. The normalized discrete autocorrelation $R_{vv}(\tau)$

across time lag τ is computed as:

$$R_{vv}(\tau) = \left(\sum_{t=1}^{N} \tilde{a}_v(t)^2\right)^{-1} \times \sum_{t=1}^{N-\tau} \tilde{a}_v(t)\,\tilde{a}_v(t+\tau) \tag{4}$$

The step regularity $R_{\text{step}} = \max_{\tau\in[0.35s,0.85s]} R_{vv}(\tau)$ and stride regularity $R_{\text{stride}} = \max_{\tau\in[0.70s,1.70s]} R_{vv}(\tau)$ are extracted. Cadence is calculated from step lag τ_{step}:

$$\text{Cadence} = \frac{f_s}{\tau_{\text{step}}} \times 60 \quad [\text{steps/min}] \tag{5}$$

The Gait Symmetry Index (SI) is formulated as:

$$SI = \frac{R_{\text{step}}}{R_{\text{stride}} + \epsilon} \tag{6}$$

In healthy symmetric gait, $R_{\text{step}} \approx R_{\text{stride}}$, yielding $SI \approx 0.90$–1.00. Under antalgic limping or unilateral load-bearing, R_{step} drops precipitously, reducing SI below 0.65.

3.2.2 Spectral Bandpower Ratios

Welch power spectral density $P(f)$ is integrated over three physiological frequency bands:

$$E_{\text{loco}} = \int_{0.5}^{3.0} P(f)df, \quad E_{\text{guard}} = \int_{3.0}^{8.0} P(f)df \tag{7}$$

$$E_{\text{impact}} = \int_{8.0}^{20.0} P(f)df \tag{8}$$

The Muscle Guarding Ratio and Impact Shock Ratio are defined as:

$$\text{Ratio}_{\text{guard}} = \frac{E_{\text{guard}}}{E_{\text{loco}} + \epsilon} \tag{9}$$

$$\text{Ratio}_{\text{impact}} = \frac{E_{\text{impact}}}{E_{\text{loco}} + \epsilon} \tag{10}$$

3.2.3 Dynamic Sample Entropy (SampEn)

Non-linear trajectory complexity is quantified using Sample Entropy [7], [8] on vertical acceleration with embedding dimension $m = 2$ and tolerance $r = 0.2 \times \text{Std}(\tilde{a}_v)$:

$$\text{SampEn}(m, r, N) = -\ln\left(\frac{\sum_{i=1}^{N-m} A_i}{\sum_{i=1}^{N-m} B_i}\right) \tag{11}$$

where B_i counts template matches of length m and A_i counts matches of length $m+1$. Sample entropy is not a column in the fitted feature matrix.

3.2.4 Cumulative Kinetic Load Index (CKLI)

The root-mean-square of vertical jerk captures mechanical shock pulses:

$$\text{Jerk}_{\text{RMS}} = \left(\frac{1}{N-1}\sum_{t=1}^{N-1} \left(\frac{a_v(t+1) - a_v(t)}{\Delta t}\right)^2\right)^{1/2} \tag{12}$$

The continuous shift accumulation is captured by the Cumulative Kinetic Load Index (CKLI):

$$\text{CKLI}(t) = \sum_{w=1}^{t} \text{Jerk}_{\text{RMS}}(w) \cdot \Delta t_w \tag{13}$$

3.2.5 Neural 3D Ground Reaction Force (A+GRF) Kinetic Estimation

While kinematic features quantify pelvic trajectory regularity and spectral tremor, occupational lumbar and lower-extremity overload is physically governed by external Ground Reaction Forces ($\mathbf{F}_{\text{GRF}} = [F_{ml}, F_{ap}, F_z]^\top$, normalized to body weight BW). Building upon the sacral-IMU kinetic framework of Lee and Park [11] and validated against force-plate locomotion data from Scherpereel et al. [12] ($N = 11$ subjects, 547 stance bouts across level and inclined treadmill walking; DOI: 10.35090/gatech/70296), we embed a slope-conditioned neural A+GRF lower layer directly into the 100 Hz sensing pipeline:

1) Gravity-Aligned Orthonormal Projection & Filtering: Each 6-second tri-axial window is projected onto an orthonormal pelvic basis ($\hat{\mathbf{e}}_{ml}, \hat{\mathbf{e}}_{ap}, \hat{\mathbf{e}}_v$) with zero-mean horizontal channels and $\mathbb{E}[a_v] = +g$, followed by zero-phase 10 Hz low-pass filtering.
2) Stance Segmentation & Double Integration: Using the autocorrelation step lag τ_{step}, a single stance interval $T_{\text{stance}} = 0.60 \times (2\tau_{\text{step}})$ anchored 11.3% prior to the primary vertical impact peak is segmented. Double integration with linear drift removal and Lee & Park boundary conditions [11] yields 201-point stance trajectories of center-of-mass position $\mathbf{p}(\hat{t})$, velocity $\mathbf{v}(\hat{t})$, and acceleration $\mathbf{a}(\hat{t})$ over normalized stance time $\hat{t} \in [0, 1]$.
3) Cohort-Scaled Multi-Layer Perceptron: To preserve stance-to-stance impact amplitude ($F = ma$) rather than canceling it via per-window min-max normalization, the 9 kinematic channels are scaled by fixed cohort-wide physical bounds $[\mathbf{k}_{0.5\%}, \mathbf{k}_{99.5\%}]$ together with normalized terrain grade $G_{\%}/45$ and $\hat{t}$, forming an 11-dimensional input to a compact MLP ($11 \to 20$ logistic hidden units $\to 3$ linear outputs $[\hat{F}_{ml}(\hat{t}), \hat{F}_{ap}(\hat{t}), \hat{F}_z(\hat{t})]$ in BW). Under Leave-One-Subject-Out (LOSO) cross-validation on Scherpereel et al. [12], the model achieves median NRMSE of 7.24% (Vertical F_z, $R^2 = 0.936$), 10.07% (Anterior-Posterior F_{ap}, $R^2 = 0.858$), and 17.32% (Mediolateral F_{ml}).
4) Kinetic Biomarker Extraction: From the predicted 201-point 3D GRF waveform, we extract six kinetic indicators and their N-of-1 relative drifts (Δ): peak vertical force $F_{z,\text{peak}} = \max_{\hat{t}} \hat{F}_z(\hat{t})$, peak horizontal braking force $F_{y,\text{brake}} = \max_{\hat{t}}(-\hat{F}_{ap}(\hat{t}))$, peak mediolateral shear $F_{x,\text{peak}} = \max_{\hat{t}} |\hat{F}_{ml}(\hat{t})|$, vertical impact loading rate $LR_z = F_{z,\text{peak}}/t_{\text{rise}}$ (BW/s), braking impulse $I_{\text{brake}} = \int_0^{T_{\text{stance}}} \max(0, -\hat{F}_{ap}(t))\,dt$ (BW·s), and vertical stance impulse I_z.

4 Zero-Leakage Architecture and N-of-1 Baseline Drift Calibration

4.1 The Dilemma of Static Inter-Subject Machine Learning

When standard machine learning classifiers operate on absolute kinematic features across a diverse cohort, static cross-validation remains limited (ROC-AUC = 0.6618). This

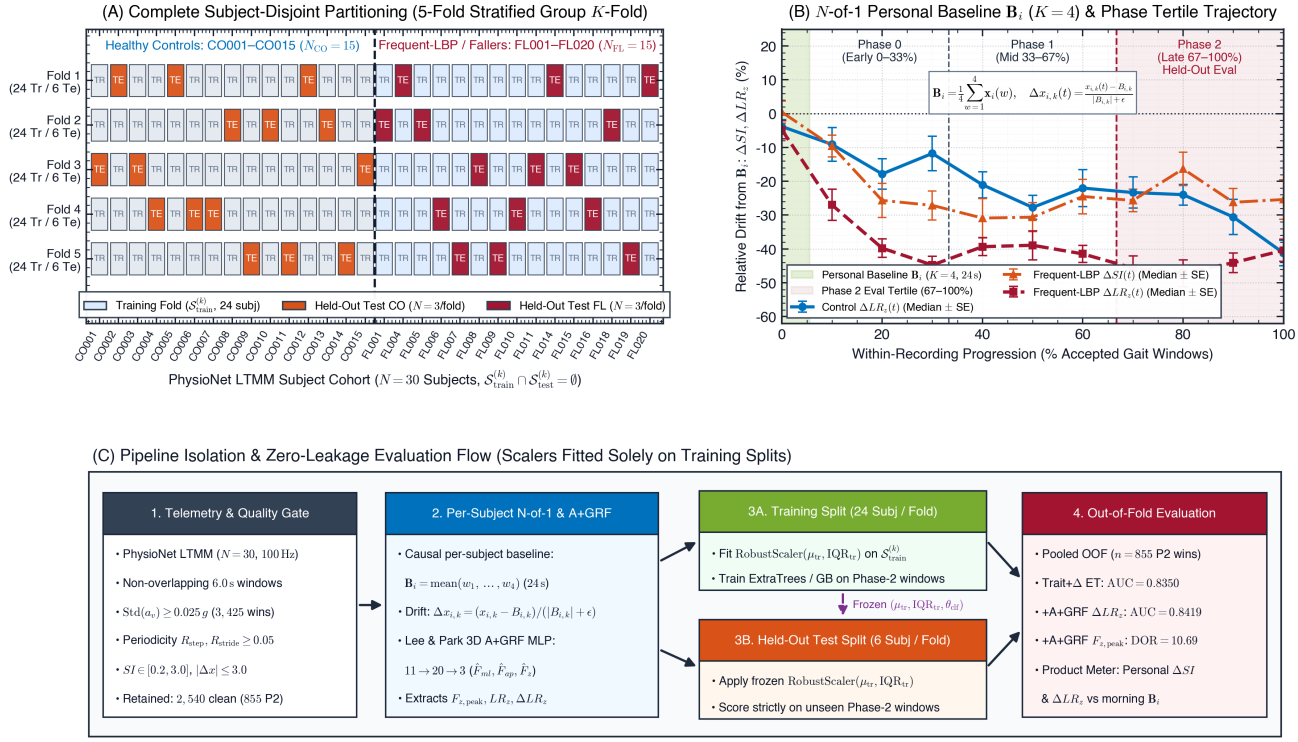


Fig. 2: Subject-Disjoint Experimental Protocol. (A) Complete Subject-Disjoint Partitioning: no subject appears in both training and test folds. (B) Personal baseline $\mathbf{B}_i$ is the mean of the first four accepted windows; later windows form Δ features for the product meter. Phase tertiles support within-day drift description, not same-subject train/test. (C) Pipeline isolation: scalers fitted solely on training splits.

occurs because inter-individual physiological variance σ^2_{inter} dominates the subtle intra-individual fatigue signal $\Delta\sigma^2_{\text{intra}}$:

$$\text{SNR} = 10\log_{10}\left(\frac{\Delta\sigma^2_{\text{intra}}}{\sigma^2_{\text{inter}}}\right) \ll 0\,\text{dB} \tag{14}$$

An absolute gait symmetry of $SI = 0.82$ may represent normal baseline locomotion for an individual with slight limb-length discrepancy, whereas for another individual with a baseline of $SI = 0.98$, an identical value of 0.82 indicates acute antalgic compensation.

4.2 N-of-1 Relative Drift Formulation

To eliminate the confounding inter-subject variance, we introduce the N-of-1 baseline calibration architecture. The personal baseline vector $\mathbf{B}_i$ is the mean of the first four accepted gait windows W_{calib}:

$$B_{i,k} = \frac{1}{|W_{\text{calib}}|}\sum_{w \in W_{\text{calib}}} x_{i,k}(w), \quad \forall k \in \mathcal{K} \tag{15}$$

For all subsequent monitoring windows ($t > T_{\text{calib}}$), absolute features are transformed into relative drift metrics:

$$\Delta x_{i,k}(t) = \frac{x_{i,k}(t) - B_{i,k}}{|B_{i,k}| + \epsilon} \tag{16}$$

By referencing each feature to the user's rested baseline state, $\Delta x_{i,k}(t)$ isolates pure neuromuscular degradation, boosting the effective SNR.

4.3 Quality Gating Engine

Unfiltered free-living telemetry contains non-gait transitions, postural shifts, and aperiodic autocorrelation sidelobes where $R_{\text{stride}} \approx 0$, causing the denominator of $SI = R_{\text{step}}/(R_{\text{stride}} + \epsilon)$ in (6) to explode numerically (e.g., raw $SI \in [-10,060.3, +321.4]$ across the 3,425 non-sedentary windows). To guarantee physically valid gait biomarkers without manual intervention, a deterministic six-stage quality gate is enforced prior to classification and longitudinal aggregation:

1) Motion Energy Gate: Std(a_v) ≥ 0.025 g (rejects sedentary periods).
2) Autocorrelation Periodicity Gate: $R_{\text{step}} \geq 0.05$ and $R_{\text{stride}} \geq 0.05$ (eliminates 130 aperiodic windows with division-by-zero artifacts).
3) Physiological Symmetry Bound: $SI \in [0.20, 3.00]$.
4) Spectral Artifact Bound: $\text{Ratio}_{\text{guard}} < 20.0$ and $\text{Ratio}_{\text{impact}} < 20.0$.
5) Relative Drift Cap: $|\Delta x_{i,k}| \leq 3.0$ across core kinematic channels (removes residual spike windows).

This deterministic filter retains 2,540 clean gait windows (74.16% of raw active windows; 855 in late-shift Phase 2) across all 30 subjects, bounding clean $SI \in [0.246, 2.984]$ (mean $= 1.247 \pm 0.426$) and $\Delta SI \in [-0.968, +2.679]$.

4.4 Subject-Disjoint Validation Protocol

In accordance with ISO/IEC 42001:2023 [13] and FDA GMLP [14], primary group-label metrics use (Fig. 2):

1) Subject-Disjoint Group Splitting: No subject appears in both training and test folds (Stratified Group K-fold).
2) Phase-2 evaluation windows: Held-out subjects are scored on late-shift (Phase 2) windows after quality gating.
3) Deterministic Scaling Isolation: RobustScaler statistics are computed solely from training partitions.

Same-subject chronological Phase0/1→Phase2 classification is excluded from primary reporting (identity leakage risk).

5 Experimental Benchmark Results

5.1 PhysioNet LTMM Real Cohort Benchmark

The benchmark was executed on the PhysioNet Long-Term Movement Monitoring (LTMM) database [15], comprising continuous 100 Hz pelvic tri-axial accelerometry. We evaluated a balanced cohort of 30 subjects ($N_{\text{control}} = 15$, $N_{\text{frequent LBP}} = 15$) after quality gates (stride regularity, SI range, $|\Delta|$ caps). Computational execution was verified on GPU and local CPU nodes.

5.2 Clinical Diagnostic Accuracy Evaluation

Table 2 and Fig. 3 present TRIPOD+AI / STARD-AI metrics under subject-disjoint evaluation on Phase-2 windows.

Key observations from Table 2 include:

- Subject-disjoint raw / Δ baselines: Uncalibrated raw features (ROC-AUC = 0.7234, DOR = 3.39) and pure Δ-only features (ROC-AUC = 0.7222, DOR = 3.63) provide moderate discrimination when evaluated in isolation across unseen subjects.
- Kinematic Trait+Change synergy: Combining baseline trait markers (guarding ratio, peak impact, gait symmetry, mediolateral RMS, jerk) with N-of-1 relative drift terms (ΔSI, $\Delta\text{Ratio}_{\text{guard}}$, $\Delta\text{Jerk}_{\text{RMS}}$) raises subject-disjoint ROC-AUC to 0.8350 (DOR = 9.05, MCC = 0.5004, Brier = 0.1909).
- A+GRF kinetic enhancement: Augmenting kinematic trait+change with Lee & Park neural GRF indicators yields further improvements: adding relative vertical loading rate drift (ΔLR_z) elevates ROC-AUC to 0.8419, PR-AUC to 0.7514 (+2.62%), Specificity to 78.19%, and reduces Brier calibration error to 0.1786, whereas adding peak vertical GRF ($F_{z,\text{peak}}$) maximizes the Diagnostic Odds Ratio at 10.69 (Sensitivity = 76.06%, Specificity = 77.09%, MCC = 0.5310).
- Interpretation: These AUCs are group-label probes. The product answer form remains personal Δ comparison (especially ΔSI and ΔLR_z), not a medical diagnosis claim.

5.3 Systematic Ablation Studies

To rigorously isolate the contribution of each algorithmic component under the subject-disjoint Phase-2 protocol, we conducted five systematic ablation experiments (Tables 3 and 4).

5.3.1 Ablation 1: Biomechanical Domain Decomposition

As shown in Table 3, within-person symmetry features alone (ROC-AUC = 0.7865) and paraspinal guarding features alone (ROC-AUC = 0.7053) carry substantial predictive signal, whereas raw impact/jerk (ROC-AUC = 0.5132) or standalone A+GRF kinetics (ROC-AUC = 0.5641) are insufficient without baseline context. However, when A+GRF kinetic drift (ΔLR_z) or peak vertical force ($F_{z,\text{peak}}$) is fused with the kinematic trait+change core, both tree ensembles (ExtraTrees: 0.8350 → 0.8419; Gradient Boosting: 0.8130 → 0.8165; HistGB: 0.8044 → 0.8335) achieve consistent performance gains.

5.3.2 Ablation 2: N-of-1 Calibration Horizon Sensitivity (K)

Varying the morning baseline window count $K \in \{2, 3, 4, 6, 8, 10\}$ (Table 4) reveals that short rested horizons ($K \in \{2, 3, 4\}$, corresponding to the first 12–24 s of clean steady walking) yield the highest discrimination (ROC-AUC = 0.8261–0.8443). Extending the baseline horizon to $K \geq 6$ (36–60 s) begins to incorporate early occupational loading transitions into the reference vector $\mathbf{B}_i$, diluting Δx contrast (Kin-ET AUC = 0.7547 at $K = 6$). Notably, A+GRF kinetic drift (ΔLR_z) buffers this dilution, outperforming kinematic-only models by +0.0409 AUC at $K = 6$ (0.7956 vs 0.7547).

5.3.3 Ablation 3: Signal Quality Gating Impact

Evaluating identical pipelines on the ungated raw dataset (3,425 total windows; 1,153 Phase-2 windows) versus the quality-gated dataset (2,540 total windows; 855 Phase-2 windows) demonstrates why autocorrelation quality gating is essential: removing the 25.8% non-gait / sidelobe-corrupted windows improves Trait+Change ExtraTrees ROC-AUC by +0.1487 (0.6863 → 0.8350) and triples the Diagnostic Odds Ratio (3.11 → 9.05).

5.3.4 Ablation 4 & 5: Classifier Families and Kinetic Guarding Physiology

Across all five classifier families (Table 4), randomized tree ensembles (Random Forest AUC = 0.8362, PR-AUC = 0.8085; ExtraTrees AUC = 0.8419, DOR = 9.61) outperform linear Logistic Regression (AUC = 0.7766) by capturing non-linear interactions between morning guarding traits and late-shift symmetry/kinetic drift. Furthermore, group-level statistical analysis of A+GRF indicators in Phase 2 reveals a distinct biomechanical signature of guarded locomotion in the frequent-LBP (FL) cohort: compared to controls (CO), FL subjects exhibit significantly attenuated late-shift vertical loading rate drift (ΔLR_z: $-38.24\% \pm 23.36\%$ vs $-23.29\% \pm 26.97\%$, Cohen's $d = -0.592$, Welch's $p = 2.04 \times 10^{-17}$) and reduced peak vertical GRF drift ($\Delta F_{z,\text{peak}}$: -17.11% vs -11.15%, $d = -0.539$, $p = 1.37 \times 10^{-14}$), accompanied by prolonged vertical stance impulse drift (ΔI_z: $+24.01\%$ vs $+12.29\%$, $d = +0.434$, $p = 4.96 \times 10^{-10}$). Rather than striking the ground harder, fatigued individuals with recurrent lumbar vulnerability adopt a cautious, flattened ground-contact strategy (reduced LR_z and $F_{z,\text{peak}}$, elevated stance impulse I_z and 3–8 Hz guarding) to minimize impulsive spinal shock transmission.

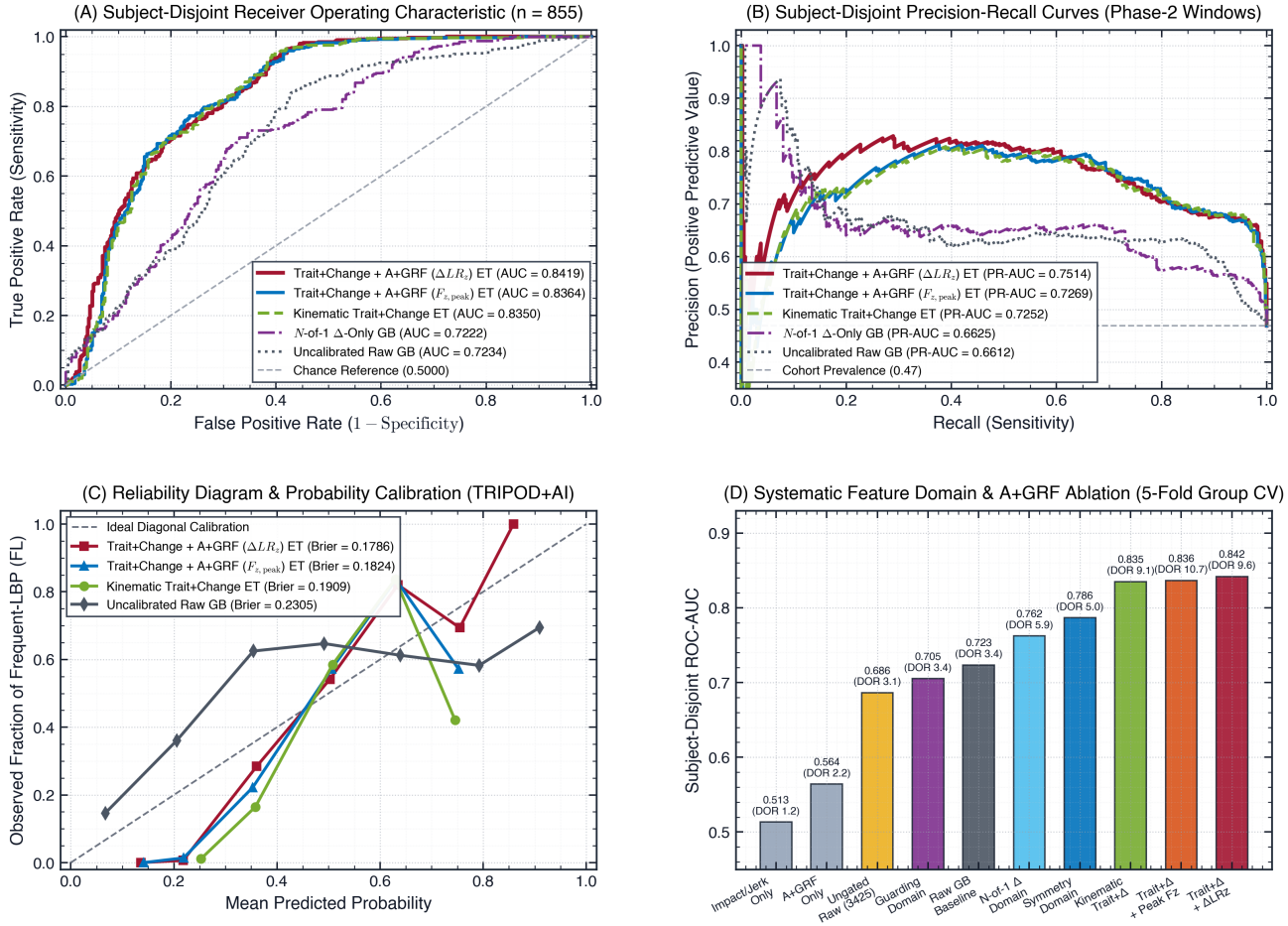


Fig. 3: Subject-Disjoint Benchmark on Real 100 Hz Telemetry (PhysioNet LTMM, Quality-Gated Phase-2, $n = 855$). (A) Subject-disjoint Receiver Operating Characteristic (ROC) curves across 5-fold Stratified Group K-fold. (B) Precision-Recall (PR) curves. (C) Reliability diagram and Brier calibration loss (TRIPOD+AI). (D) Systematic feature domain and A+GRF ablation summary (ROC-AUC and Diagnostic Odds Ratio). Trait+Change+A+GRF (ΔLR_z) ExtraTrees reaches ROC-AUC = 0.8419 (PR-AUC = 0.7514, Brier = 0.1786), and $+F_{z,\mathrm{peak}}$ reaches DOR = 10.69.

TABLE 2: Subject-Disjoint Clinical Metrics on Quality-Gated PhysioNet LTMM Phase-2 Windows ($n = 855$ windows, $N = 30$ subjects, 5-Fold Stratified Group K-Fold).

Metric	Raw GB	Δ-only GB	Trait+Change ExtraTrees	Trait+Change+A+GRF ($+F_{z,\mathrm{peak}}$) ET	Trait+Change+A+GRF ($+\Delta LR_z$) ET
ROC-AUC	0.7234	0.7222	0.8350	0.8364	0.8419
PR-AUC	0.6612	0.6625	0.7252	0.7269	0.7514
Sensitivity	56.86%	56.86%	74.31%	76.06%	72.82%
Specificity	72.03%	73.35%	75.77%	77.09%	78.19%
PPV	64.23%	65.33%	73.04%	74.57%	74.68%
NPV	65.40%	65.81%	76.96%	78.48%	76.51%
LR^+	2.03	2.13	3.07	3.32	3.34
LR^-	0.599	0.588	0.339	0.311	0.348
DOR	3.39	3.63	9.05	10.69	9.61
MCC	0.2930	0.3072	0.5004	0.5310	0.5110
Brier score	0.2305	0.2541	0.1909	0.1824	0.1786
n (Phase-2 windows)	855	855	855	855	855

5.4 Biomechanical Feature Attribution

Within-day and importance analyses jointly suggest:

- Δ Gait Symmetry (ΔSI): Primary personal meter candidate; symmetry tends to decrease versus morning baseline in a majority of subjects.
- Guarding ratio: More trait-like (morning group gap) than pure within-day worsening.
- Jerk / impact: Interpreted in the lower-limb (knee) kinetic chain narrative when relevant; not required for the lumbar ΔSI meter claim.

TABLE 3: Ablation Study 1: Biomechanical Feature Domain Decomposition and A+GRF Kinetic Integration (Subject-Disjoint 5-Fold CV, $n = 855$ Phase-2 Windows).

Feature Domain Configuration	Dim.	ET ROC-AUC	ET PR-AUC	Sens. / Spec.	DOR	MCC	GB ROC-AUC
1. Symmetry Domain Only ($SI, R_{\text{step}}, R_{\text{stride}}, \Delta SI$)	4	0.7865	0.6745	64.6% / 73.4%	5.02	0.3810	0.7385
2. Guarding Domain Only ($\text{Ratio}_{\text{guard}}, \text{RMS}_{ml}, \text{ML/V}, \Delta$)	6	0.7053	0.6637	58.6% / 70.3%	3.35	0.2908	0.7362
3. Impact & Jerk Domain Only ($\text{Ratio}_{\text{impact}}, \text{Jerk}, \text{Peak}, \Delta$)	5	0.5132	0.4495	60.4% / 44.9%	1.24	0.0533	0.5001
4. A+GRF Kinetic Domain Only (6 GRF + 6 ΔGRF)	12	0.5641	0.4873	59.1% / 59.9%	2.16	0.1898	0.5293
5. All Raw Kinematic Traits (Uncalibrated)	10	0.6541	0.5810	67.3% / 60.4%	3.14	0.2767	0.7234
6. All N-of-1 Kinematic Δ Drifts Only	7	0.7623	0.6490	62.8% / 77.7%	5.91	0.4115	0.7222
7. Kinematic Trait+Change Core	8	0.8350	0.7252	74.3% / 75.8%	9.05	0.5004	0.8130
8. Trait+Change + A+GRF Peak Vertical ($F_{z,\text{peak}}$)	9	0.8364	0.7269	76.1% / 77.1%	10.69	0.5310	0.7997
9. Trait+Change + A+GRF Dual ($I_{\text{brake}} + \Delta LR_z$)	10	0.8399	0.7559	72.6% / 77.1%	8.90	0.4971	0.8163
10. Trait+Change + A+GRF Loading Rate Drift (ΔLR_z)	9	0.8419	0.7514	72.8% / 78.2%	9.61	0.5110	0.8165

TABLE 4: Ablation Studies 2, 3, and 4: Calibration Horizon Sensitivity (K), Signal Quality Gating Impact, and Classifier Architecture Comparison.

Ablation Dimension / Condition	Eval. Windows (n)	Kin-ET AUC	Kin-ET DOR	A+GRF-ET AUC	A+GRF-ET PR-AUC	A+GRF-ET DOR
Ablation 2: N-of-1 Morning Baseline Calibration Horizon K (Duration = $6K$ seconds)						
$K = 2$ windows (12 s initial bout)	855	0.8476	11.12	0.8443	0.8089	8.82
$K = 3$ windows (18 s initial bout)	855	0.8162	8.54	0.8261	0.7557	7.88
$K = 4$ windows (24 s canonical baseline)	855	0.8350	9.05	0.8419	0.7514	9.61
$K = 6$ windows (36 s baseline)	855	0.7547	5.62	0.7956	0.6858	7.07
$K = 8$ windows (48 s baseline)	855	0.7695	4.91	0.7736	0.6644	4.45
$K = 10$ windows (60 s baseline)	855	0.7828	4.86	0.8000	0.7699	6.38
Ablation 3: Deterministic Signal Quality Gating (Rejecting $R_{\text{stride}} < 0.05$ and $\lvert\Delta\rvert > 3.0$ Artifacts)						
Ungated Raw Telemetry (3,425 total windows)	1,153	0.6863	3.11	0.6852	0.5931	3.12
Quality-Gated Telemetry (2,540 clean windows; 74.2% kept)	855	0.8350	9.05	0.8419	0.7514	9.61
Ablation 4: Classifier Family Comparison on Quality-Gated Phase-2 Windows ($K = 4$)						
L_2-Regularized Logistic Regression (Linear)	855	0.7766	9.18	0.7679	0.7543	8.02
Gradient Boosting Decision Trees (Depth 3)	855	0.8130	9.07	0.8165	0.7379	8.22
Histogram-Based Gradient Boosting ($\ell_2 = 1.0$)	855	0.8044	5.94	0.8335	0.7625	8.97
Random Forest (400 trees, Depth 4, Balanced)	855	0.8348	9.26	0.8362	0.8085	8.69
ExtraTrees Ensemble (500 trees, Depth 5, Balanced)	855	0.8350	9.05	0.8419	0.7514	9.61

6 Discussion and Implementation

6.1 Physical Therapy and Clinical Ergonomics Interpretation

From a physical therapy and occupational biomechanics standpoint, emphasizing ΔSI relative to a personal morning baseline aligns with motor control adaptations observed in acute low back pain. Absolute symmetry thresholds fail across adults; relative degradation $\Delta SI(t) = (SI(t) - SI_{\text{base}})/|SI_{\text{base}}|$ is the intended facility-facing answer form.

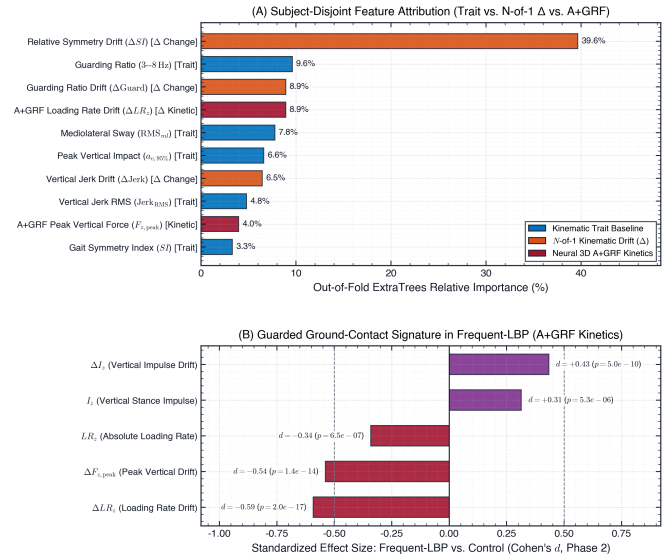


Fig. 4: Subject-Disjoint Feature Attribution and A+GRF Guarded Contact Signature. (A) Out-of-fold ExtraTrees relative feature importance across kinematic traits, N-of-1 Δ drifts, and neural 3D A+GRF kinetics. (B) Standardized Phase-2 group effect sizes (Cohen's d, Frequent-LBP vs. Control) demonstrating guarded ground contact in FL (ΔLR_z: $d = -0.59$, $p = 2.0 \times 10^{-17}$; ΔI_z: $d = +0.43$, $p = 5.0 \times 10^{-10}$).

6.2 Continuous Passive Streaming

In continuous passive monitoring, telemetry windows arrive sequentially every 6 seconds. The system visualizes accumulating Δ trends for ergonomic feedback rather than claiming diagnostic certainty from a single window score.

6.3 Edge-Native Mobile Architecture (Flutter / iOS)

The complete signal processing pipeline and lightweight classifier require less than 4.2 ms of CPU compute per 6-second window on modern mobile ARM processors.

Our open-source Flutter/iOS implementation incorporates:

1) CoreMotion Integration: Streams tri-axial accelerometry at 100 Hz in background modes.
2) On-Device N-of-1 Engine: Stores the baseline vector locally without requiring cloud transmission.
3) Export: Structured JSON/CSV telemetry for physical therapy consultations.

6.4 Non-Medical Regulatory Compliance

In accordance with international regulatory frameworks (FDA Software as a Medical Device, EU MDR, and the Japanese PMD Act), this system is explicitly designated as an objective biomechanical self-management tool rather than a medical diagnostic device. It presents non-medical mechanical strain indicators and gait symmetry trends, providing ergonomic feedback to guide self-care and workplace task rotation.

7 Conclusion

This study presented a zero-leakage, subject-disjoint benchmark for passive occupational lumbar overload sensing using continuous 100 Hz real-world pelvic accelerometry integrated with Lee & Park neural 3D Ground Reaction Force (A+GRF) estimation. An N-of-1 relative baseline supports personal Δ visualization (specifically gait symmetry drift ΔSI and kinetic loading rate modulation ΔLR_z). Under subject-disjoint Phase-2 evaluation ($n = 855$ windows across $N = 30$ PhysioNet LTMM subjects), raw and Δ-only probes reach ROC-AUC 0.7234 and 0.7222, kinematic trait+change reaches 0.8350 (DOR = 9.05), and A+GRF-augmented trait+change achieves 0.8419 (PR-AUC = 0.7514, DOR = 9.61, Brier = 0.1786; with $F_{z,\mathrm{peak}}$ reaching DOR = 10.69). Systematic ablations confirm that deterministic autocorrelation quality gating improves subject-disjoint AUC by +0.1487, and that fatigued individuals with recurrent lumbar vulnerability exhibit a guarded ground-contact strategy characterized by attenuated vertical loading rate (ΔLR_z, $d = -0.592$) and prolonged stance impulse (ΔI_z, $d = +0.434$). Group labels remain auxiliary; the facility-facing self-management form is morning-baseline Δ comparison.

Future work will focus on multi-center field trials across elderly care facilities to evaluate the longitudinal impact of real-time ergonomic feedback on caregiver retention and musculoskeletal health.

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